LABSHEET 1

1. Create a database named CollegeDB.

```
CREATE DATABASE CollegeDB;
```

2. Create a Student table with Student_ID, Name, Age, Course, and City.

```
CREATE TABLE Student (
 Student_ID INT,
 Name VARCHAR(100),
 Age INT,
 Course VARCHAR(100),
 City VARCHAR(100)
);
```

3. Create an Employee table with Emp_ID, Emp_Name, Department, and Salary.

```
CREATE TABLE Employee (
 Emp_ID INT,
 Emp_Name VARCHAR(100),
 Department VARCHAR(100),
 Salary DECIMAL(10,2)
);
```

4. Create a Department table with Dept_ID, Dept_Name, and Location.

```
CREATE TABLE Department (
 Dept_ID INT,
 Dept_Name VARCHAR(100),
 Location VARCHAR(100)
);
```

5. Create a Product table with Product_ID, Product_Name, Price, and Quantity.

```
CREATE TABLE Product (
 Product_ID INT,
 Product_Name VARCHAR(100),
 Price DECIMAL(10,2),
 Quantity INT
);
```

6. Create a Customer table with Customer_ID, Customer_Name, Email, and Phone.

```
CREATE TABLE Customer (
 Customer_ID INT,
 Customer_Name VARCHAR(100),
 Email VARCHAR(150),
 Phone VARCHAR(20)
);
```

7. Create a Course table with Course_ID, Course_Name, Duration, and Fees.

```
CREATE TABLE Course (
 Course_ID INT,
 Course_Name VARCHAR(100),
 Duration VARCHAR(50),
```

```
 Fees DECIMAL(10,2)
);
```

8. Create a Faculty table with Faculty_ID, Faculty_Name, Subject, and Salary.

```
CREATE TABLE Faculty (
 Faculty_ID INT,
 Faculty_Name VARCHAR(100),
 Subject VARCHAR(100),
 Salary DECIMAL(10,2)
);
```

9. Create a Library table with Book_ID, Book_Name, Author, and Price.

```
CREATE TABLE Library (
 Book_ID INT,
 Book_Name VARCHAR(150),
 Author VARCHAR(100),
 Price DECIMAL(10,2)
);
```

10. Create a Department table with Dept_ID as the PRIMARY KEY.

```
CREATE TABLE Department (
 Dept_ID INT PRIMARY KEY,
 Dept_Name VARCHAR(100),
 Location VARCHAR(100)
);
```

11. Create a Student table with Student_ID as PRIMARY KEY and Name as NOT NULL.

```
CREATE TABLE Student (
 Student_ID INT PRIMARY KEY,
 Name VARCHAR(100) NOT NULL
);
```

12. Create an Employee table with a UNIQUE constraint on Email.

```
CREATE TABLE Employee (
 Emp_ID INT PRIMARY KEY,
 Emp_Name VARCHAR(100),
 Email VARCHAR(150) UNIQUE
);
```

13. Create a Product table with a CHECK constraint that Price > 0.

```
CREATE TABLE Product (
 Product_ID INT PRIMARY KEY,
 Product_Name VARCHAR(100),
 Price DECIMAL(10,2) CHECK (Price > 0),
 Quantity INT
);
```

14. Add a new Email column to the Student table using ALTER TABLE.

```
ALTER TABLE Student ADD COLUMN Email VARCHAR(150);
```

15. Add a Phone column to the Student table.

```
ALTER TABLE Student ADD COLUMN Phone VARCHAR(20);
```

16. Modify the datatype/size of the Name column in the Student table.

```
ALTER TABLE Student MODIFY COLUMN Name VARCHAR(150);
```

17. Rename the Student table to Student_Details.

```
RENAME TABLE Student TO Student_Details;
```

18. Rename the column City to Address.

```
ALTER TABLE Student CHANGE COLUMN City Address VARCHAR(100);
```

19. Add a Salary column to the Employee table.

```
ALTER TABLE Employee ADD COLUMN Salary DECIMAL(10,2);
```

20. Drop the Phone column from the Student table.

```
ALTER TABLE Student DROP COLUMN Phone;
```

21. Drop the Email column from the Student table.

```
ALTER TABLE Student DROP COLUMN Email;
```

22. Remove all records from a table using TRUNCATE.

```
TRUNCATE TABLE Student;
```

23. Drop the Course table from the database.

```
DROP TABLE Course;
```

24. Drop the Customer table from the database.

```
DROP TABLE Customer;
```

25. Create a complete Student table using PRIMARY KEY, NOT NULL, UNIQUE and CHECK constraints, and demonstrate suitable ALTER and DROP operations.

```
CREATE TABLE Student (
 Student_ID INT PRIMARY KEY,
 Name VARCHAR(100) NOT NULL,
 Email VARCHAR(150) UNIQUE,
 Age INT CHECK (Age >= 16),
 Course VARCHAR(100),
 City VARCHAR(100)
);

ALTER TABLE Student ADD COLUMN Phone VARCHAR(20);
ALTER TABLE Student MODIFY COLUMN Name VARCHAR(150);
ALTER TABLE Student CHANGE COLUMN City Address VARCHAR(100);
ALTER TABLE Student DROP COLUMN Phone;
DROP TABLE Student;
```